Applications of the Polynomial s-Power Basis in Geometry Processing

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We propose a unified methodology to tackle geometry processing operations admitting explicit algebraic expressions. This new approach is based on representing and manipulating polynomials algebraically in a recently presented basis, the symmetric analogue of the power form (s-power basis for brevity), so called because it is associated with a "Hermite two-point expansion" instead of a Taylor expansion. Given the expression of a polynomial in this basis over the unit interval $u \in [0, 1]$, degree reduction is trivially obtained by truncation, which yields the Hermite interpolant that matches the original derivatives at $u = \{0, 1\}$. Operations such as division or square root become meaningful and amenable in this basis, since we can compute as many terms as desired of the corresponding Hermite interpolant and build "s-power series," akin to Taylor series. Applications include computing integral approximations of rational polynomials, or approximations of offset curves.

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1. INTRODUCTION

1.1 Processing Polynomials in CAGD

Geometry processing is defined as the calculation of geometric properties of already constructed geometric objects [Barnhill 1992]. In its most comprehensive meaning, this term includes all the algorithms that are applied to

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already existing geometric entities [Farin 1989]. Thus, this topic includes the computation of curve lengths, surface areas, offset curves and surfaces, pipe or sweep surfaces from a spine curve, or conversions between different curve and surface forms. At first glance, many of these computations seem quite straightforward because they admit exact explicit algebraic expressions involving just integrals, derivatives, and standard arithmetical operations, such as addition, multiplication, division, and square roots. The actual difficulty is that these operations involve polynomial expressions in the customary case of polynomial curves and surfaces.

Adding two polynomials of respective degrees n_1, n_2 is in principle a simple task, as this operation, which results in a polynomial of degree $n = \max(n_1, n_2)$, reduces to component-wise addition given a polynomial basis. However, if $n_1 \neq n_2$, this operation is not so trivial in the standard Bernstein basis, since the polynomial of lowest degree must be degree-raised to degree n in order to obtain a common representation. Multiplication is more troublesome because the resulting polynomial has degree $n_1 + n_2$. Thus, if we perform subsequent multiplications, the degree may become prohibitively high or out of the range our computer system can deal with, and we must resort to degree-reduction to represent the result as a polynomial of reasonable degree. Composing two polynomial functions is much worse, since it yields a polynomial of degree $n_1 n_2$. Nevertheless, by far the toughest operations are division and square root, whose result in general can not be expressed exactly as a polynomial. Stated formally, the set of polynomials is not closed with respect to these operations. Therefore, the only way of tackling a division or a square root is to find a suitable approximation.

Little systematic research has been undertaken in this area, giving rise to a multitude of mostly ad hoc algorithms to solve each particular geometry processing problem. In summary, in this area there is neither a unified approach nor "key developments" such as the Bézier technique [Piegl 1989]. This is why Farin [1989;1997] contends that "at present, no unified approach exists for these geometry processing problems" and that, in consequence, "future research should not only aim at the development of new algorithms, but also at the unification of methodology".

1.2 The Power Basis

It might seem that the use of the monomial (power) basis $\{1, u, u^2, \ldots, u^n\}$ provides us with a solution to the problem, since in this basis:

- The actual degree of a polynomial is immediately apparent by inspection of its coefficients.
- If a polynomial has a degree too large, we may simply drop the high-degree terms to obtain an approximation.
- Addition, multiplication, differentiation, and antidifferentiation are grade school exercises.

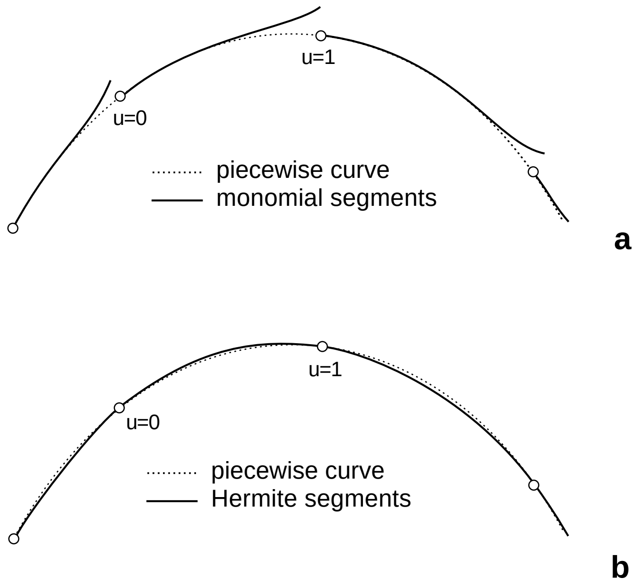

Fig. 1. Piecewise approximations: (a) Taylor and (b) Hermite.

- There exist algorithms to compute a polynomial approximation in power form for the quotient of two polynomials, or the square root of a polynomial.

In fact, we can find a polynomial approximation (a power, i.e., Taylor series) for most important functions. Thus the polynomial power form is naturally embedded in the domain of power series, which are dealt with in most aspects like polynomials [Geddes et al. 1992]. In consequence, we only need a simple computer algebra system to carry out in this basis all required mathematical operations in a geometry processing problem, and to trim the result to the desired degree. Unfortunately, while a Taylor expansion is good at approximating a function $a(u)$ over a symmetric interval around $u = 0$, it leads to gaps when we try to connect several expansions (Figure 1(a)). This is unacceptable for CAGD, as curves and surfaces are mainly defined in a piecewise manner. In addition, the power basis displays poor numerical properties.

1.3 The Proposed Unified Approach

Rather than Taylor approximations of a given function $a(u)$, the natural approach seems to employ two-point Hermite approximations. If the interval of interest is $u \in [0, 1]$, then the order-k Hermite interpolant is the unique degree-$(2k + 1)$ polynomial that reproduces the derivatives $\{a^{(r)}(0),\ a^{(r)}(1)\}_{r=0}^{k}$, in other words, a polynomial with contact of order k at the endpoints $u = \{0, 1\}$ with the function $a(u)$. Such Hermite approximations are very convenient for a piecewise representation (Figure 1(b)), since by piecing together several approximations we get a spline polynomial enjoying C^k continuity at the joints. However, there is a drawback: standard order-k Hermite polynomials are not a subset of those of higher order. Formally, the Hermite representation does not enjoy the permanence property [Davis 1975]. Hence, dealing with polynomials of different degrees is cumbersome, and no Hermite expansion is achieved.

An alternative polynomial basis, introduced in Sánchez-Reyes [1997], solves this problem. This new basis (s-power basis) can be regarded as the symmetric analogue of the power basis, because it is associated with a "Hermite expansion" over the interval $u \in [0, 1]$ instead of a Taylor expansion around $u = 0$. A truncation at the kth term of a polynomial $a(u)$ yields the order-k Hermite interpolant of $a(u)$, in other words, this truncation preserves as many derivatives at the endpoints as possible. Therefore, degree reduction is simply dropping the desired high degree terms. Regarding numerical properties, Sánchez-Reyes [1997] demonstrates that this basis displays good numerical condition. In addition, the conversion matrices between the Bernstein and s-power basis are not ill-conditioned, so that the basis conversion does not imply a severe loss of accuracy. Thus we could transform the input data to this new basis, carry out all needed operations in it, trim to the desired degree, and bring the result back to the Bernstein basis if required.

To make this promising approach feasible, we must develop algorithms to carry out common arithmetic operations in the s-power basis. This is done in Section 3 after a brief review of the polynomial basis in Section 2. Next, Section 4 explores s-power series, the two-point analogue of the Taylor series, and their convergence. Section 5 presents two applications in the geometry processing field: finding integral approximations of rational curves and approximations of offset curves. Section 6 outlines how to extend this work to bivariate polynomials. Finally, conclusions are drawn in Section 7.

2. THE SYMMETRIC POWER FORM OF A POLYNOMIAL

2.1 The s-Power Basis

In this section the symmetric power basis [Sánchez-Reyes 1997] and its properties are briefly reviewed to make the article is as self-contained as possible. Throughout this paper we use the odd-degree scaled basis (with no binomial coefficients) because it is best suited for algebraic manipulation purposes. We refer to this basis as the s-power basis. Given an integer $p \geq 1$, the basis functions of order p (odd-degree $n = 2p + 1$) are

$$\{s^k \cdot (1 - u), s^k u\}_{k=0}^{p}, \quad s = (1 - u)u. \tag{1}$$

Note that $s^k = (1 - u)^k u^k$ is nothing else than the scaled central Bernstein polynomial $B_k^{2k}(u)$ of degree $2k$. The function s^k has the following important properties:

- it is symmetric with respect to the center point $u = 1/2$, where the maximum 4^{-k} is attained;
- it is a $(k - 1)$-fold zero at $u = \{0, 1\}$, i. e., all its derivatives of order $< k$ vanish at the endpoints.

The basis functions (1) are, over $u \in [0, 1]$, the true analogue of the monomial basis, in the following sense: for $k > 0$, whereas u^k generates the linear space of degree-k polynomials that are $(k - 1)$-fold zeros at $u = 0$, the pair $\{s^k(1 - u), s^k u\}$ spawns the space of degree-$(2k + 1)$ polynomials that are $(k - 1)$-fold zeros at $u = \{0, 1\}$.

2.2 Expression of a Polynomial in the Basis

A polynomial $a(u)$ in the s-power basis takes the form

$$a(u) = \sum_{k=0}^{p} [(1 - u)a_k^0 + ua_k^1]s^k, \quad s = (1 - u)u$$

where $\{a_k^0, a_k^1\}_{k=0}^p$ denote the $n + 1$ coefficients of $a(u)$. The summation above admits two equivalent interpretations:

- A convex combination of two polynomials $a^0(s)$, $a^1(s)$ in monomial form in s:

$$a(u) = (1 - u)a^0(s) + a^1(s), \quad \left\{ \begin{array}{l} a^0(s) = \sum a_k^0 s^k \\ a^1(s) = \sum a_k^1 s^k \end{array} \right. \tag{2}$$

- A polynomial in monomial form in s, whose coefficients are not constants but linear functions $a_k(u)$:

$$a(u) = \sum_{k=0}^{p} a_k(u)s^k, \quad a_k(u) = (1 - u)a_k^0 + ua_k^1 = \{(1 - u), u\} \left\{ \begin{array}{c} a_k^0 \\ a_k^1 \end{array} \right\}. \tag{3}$$

Henceforth, the linear function $a_k(u)$ is written simply $a_k(u)$, and often referred to as a "coefficient," although always bearing in mind that it is not a constant value. Expression (3) also suggests that a natural representation of $a_k(u)$ is the following pair:

$$a_k(u) \rightarrow a_k = \left\{ \begin{array}{c} a_k^0 \\ a_k^1 \end{array} \right\}. \tag{4}$$

This pair $a_k^{0/1}$ denotes the coefficients in the Bernstein basis of the linear function $a_k(u)$, that is, its Bézier ordinates. The simplified notation (4) is adopted hereafter to represent linear functions. Observe that in case $a_k^0 = a_k^1$ the linear function simplifies to the constant $a_k = a_k^0 = a_k^1$. Thus, if

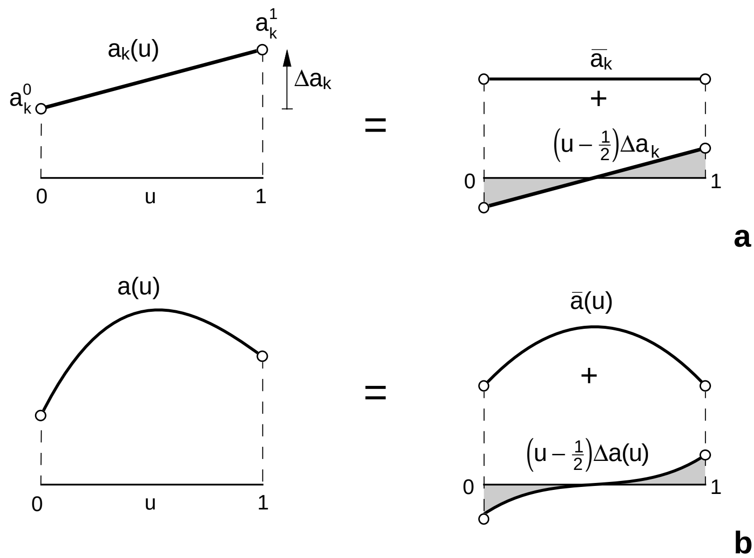

Fig. 2. Decomposing a polynomial into its symmetric and skew-symmetric components: (a) linear function $a_k(u)$; (b) cubic polynomial $a(u)$.

$a_p^0 = a_p^1$ for the last coefficient in the summation (3), then the polynomial $a(u)$ has (at most) even-degree $2p$.

2.3 Decomposition into Symmetric and Skew-Symmetric Components

In this section we present an alternative way of looking at the s-power basis and introduce some notation that will prove useful for later constructions. The key idea is to decompose the linear function $a_k(u)$ (3) into a constant term (its mean value $\bar{a}_k$) and a linear term (depending on its derivative Δa_k) (see Figure 2(a)):

$$a_k(u) = \bar{a}_k + \left(u - \frac{1}{2}\right)\Delta a_k \quad \left\{ \begin{aligned} \bar{a}_k &= \frac{1}{2}(a_k^0 + a_k^1) \\ \Delta a_k &= a_k^1 - a_k^0 \end{aligned} \right.$$

Equivalently, we employ the basis $\{1, u - 1/2\}$ rather than the Bernstein basis. For the sake of completeness, we include the expression of the Bézier ordinates $a_k^{0/1}$ in terms of $\{\bar{a}_k, \Delta a_k\}$:

$$\begin{Bmatrix} a_k^0 \\ a_k^1 \end{Bmatrix} = \bar{a}_k \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + \frac{\Delta a_k}{2} \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}. \tag{5}$$

Introducing these new coefficients in expression (3), a polynomial $a(u)$ admits the following decomposition into its symmetric and skew-symmetric components:

$$a(u) = \bar{a}(s) + \left(u - \frac{1}{2}\right)\Delta a(s) \quad \begin{cases} \bar{a}(s) = \sum_{k=0}^{p} \bar{a}_k s^k \\ \Delta a(s) = \sum_{k=0}^{p} \Delta a_k s^k \end{cases}$$

Figure 2(b) illustrates this decomposition for the case of a cubic polynomial.

By inverting the sign of the skew-symmetric part, which is tantamount to swapping coefficients $a_k^{0/1}$, we get the specular reflection $a(1 - u)$ with respect to the axis $u = 1/2$. If we now multiply $a(u)$ by its reflection, we obviously obtain a symmetric polynomial $\hat{a}(s)$:

$$\hat{a}(s) = a(u)a(1 - u) = a^0(s)a^1(s) + s(\Delta a(s))^2. \tag{6}$$

This polynomial has no skew-symmetric part, and therefore it is a true polynomial in power form in s, whose coefficients $\hat{a}_k$ are constants instead of linear functions.

2.4 Degree-Related Operations

As stressed in Sánchez-Reyes [1997], the chief advantage of utilizing the s-power basis is that all degree-related operations become trivial:

- The actual (minimum) degree $n_{\min}$ of a polynomial is simply the odd value $2p + 1$, where a_p is the nonzero pair (4) with highest subscript, except when $\Delta a_p = 0$. In this case the function a_p is the constant $a_p^0 = a_p^1$, and hence $n_{\min}$ reduces to the even value $2p$.
- Degree elevation reduces to adding zero coefficients.
- Degree reduction is obtained by truncating the summation (3) at the kth term. This operation yields the order-k Hermite interpolant $H_k(a;u)$ of $a(u)$:

$$H_k(a;u) = \sum_{i=0}^{k} a_i(u)s^i.$$

3. ALGEBRAIC MANIPULATION IN THE S-POWER FORM

3.1 Handling Hermite Two-Point Interpolants

This chapter develops the methodology to handle polynomials in the s-power basis; that is, how to deal with Hermite two-point interpolants. This topic includes algorithms to add, multiply, compose, differentiate, integrate, divide or extract square roots of polynomials. For the case of division and square root, the algorithms presented yield as many terms as desired of the Hermite interpolant of the solution. Most rules on handling Taylor approximations apply to such a Hermite two-point expansion, because in essence the latter can be thought of as a Taylor expansion in which the coefficients are linear functions. In particular, given two arbitrary functions $a(u)$, $b(u)$, in order to compute the order-k Hermite interpolant of their sum, product, or quotient, or composition, only the order-k Hermite interpolants of $a(u)$, $b(u)$ are relevant, as readily deduced from the rules on differentiation.

Given an arbitrary function $a(u)$ over the interval [0,1], we do not address the problem of how to compute its order-k Hermite interpolant in terms of its derivatives at $u = \{0, 1\}$. The interested reader can find an explicit formula in Davis [1975, p. 37]. Such a formula is not needed in this paper, except for the very simple case of the order-0 Hermite interpolant $H_0(a;u)$, which reduces to the linear function interpolating $a(u)$ at the endpoints $u = \{0, 1\}$. Using the symbolism in (4), this function is represented as

$$H_0(a;u) \rightarrow a_0 = \begin{Bmatrix} a(0) \\ a(1) \end{Bmatrix}. \tag{7}$$

3.2 Addition and Multiplication

Adding two polynomials in the s-power basis is simply a component-wise addition, as with the standard power form, although now each coefficient is a pair (4). In contrast to the Bernstein basis, there is no problem in adding two polynomials of different degrees.

Regarding multiplication, we can check that multiplying polynomials in the s-power basis is not more difficult than in the power basis. Recall that the latter is performed using the familiar "Cauchy product rule": given two polynomials in power form of degrees m, n and coefficient lists $\mathbf{a} = \{a_0, \ldots, a_m\}$, $\mathbf{b} = \{b_0, \ldots, b_n\}$, respectively, their product has degree $m + n$ and a coefficient list $\mathbf{c}$ obtained as the discrete convolution:

$$\mathbf{c} = \mathbf{a}^*\mathbf{b} = \sum_{j=0}^{n} \mathrm{shift}_j(b_j\mathbf{a}) \tag{8}$$

where the operation shift_j means shifting a list j positions to the right and filling the leading gaps with zeros. The code for the convolution (8) follows:

```
c = 0
for j := 0 to n do
   for i := j to m + j do
      c_i := c_i + b_j a_{i-j}
   endfor
endfor
```

This algorithm has a cost of $(m + 1)(n + 1)$ multiplications and additions. Note that, if a degree-k approximation suffices, then the upper limit $m + j$ in the inner loop must be replaced by $\min(k, m + j)$.

To multiply in the symmetric power form, we first have to learn how to multiply two "coefficients" (4), that is, two linear functions $a(u)$, $b(u)$ represented by their Bézier ordinates:

$$a(u) \to a = \begin{Bmatrix} a^0 \\ a^1 \end{Bmatrix}, \quad b(u) \to b = \begin{Bmatrix} b^0 \\ b^1 \end{Bmatrix}.$$

Sticking to this notation, some elementary algebra yields

$$c(u) = a(u)b(u) \to c = \begin{Bmatrix} a^0b^0 \\ a^1b^1 \end{Bmatrix} - s\Delta a \Delta b \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \quad \begin{matrix} \Delta a = a^1 - a^0 \\ \Delta b = b^1 - b^0 \end{matrix} \tag{9}$$

Thus, in addition to the linear term $H_0(c;u)$, computed by a componentwise product, there appears a quadratic term, a constant by $s = (1 - u)u$.

An order-p polynomial $a(u)$ in s-power form (2) can be represented by a pair $a^{0/1}(s)$ of polynomials in monomial form, each of them represented by their respective coefficient lists:

$$a(u) \to \mathbf{a} = \begin{Bmatrix} \mathbf{a^0} \\ \mathbf{a^1} \end{Bmatrix} = \begin{Bmatrix} \{a_0^0, \ldots, a_p^0\} \\ \{a_0^1, \ldots, a_p^1\} \end{Bmatrix}.$$

On the other hand, the product $c(u) = a(u)b(u)$ of two polynomials in s-power form (3) is computed as a summation of products $(a_i s^i)(b_j s^j)$. According to rule (9), such products can be rewritten as

$$(a_i s^i)(b_j s^j) = s^{i+j} \begin{Bmatrix} a_i^0 b_j^0 \\ a_i^1 b_j^1 \end{Bmatrix} - s^{i+j+1} \Delta a_i \Delta b_j \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}.$$

Therefore, the list $\mathbf{c}$ for the product is obtained as a function of the lists $\mathbf{a}$, $\mathbf{b}$ by replacing in expression (9) the following terms:

coefficient	→	coefficient list
product of coefficients	→	convolution* of lists
multiplication by s	→	shift to the right a list

$$c(u) \to \mathbf{c} = \begin{Bmatrix} \mathbf{c}^0 \\ \mathbf{c}^1 \end{Bmatrix} = \begin{Bmatrix} \mathbf{a}^0 * \mathbf{b}^0 - \text{shift}_1(\Delta \mathbf{a} * \Delta b) \\ \mathbf{a}^1 * \mathbf{b}^1 - \text{shift}_1(\Delta \mathbf{a} * \Delta \mathbf{b}) \end{Bmatrix}, \quad \begin{matrix} \Delta \mathbf{a} = \mathbf{a}^1 - \mathbf{a}^0 \\ \Delta \mathbf{b} = \mathbf{b}^1 - \mathbf{b}^0 \end{matrix}$$

The operation shift_1 moves the coefficients of a list $\mathbf{d}$ one position to the right and fills the leading gap with a zero:

$$\text{shift}_1(\{d_0, \ldots, d_q\}) = \{0, d_0, \ldots, d_q\}$$

Multiplication in the s-power basis has a cost similar to that in the monomial basis. Rather than convolving two lists, we now carry out 3 convolutions , but this is counterbalanced by the fact that the lists involved are half the length.

Finally, multiplication provides us with an alternative to transformation matrices for conversion from the Bernstein to the s-power basis. Given a polynomial in the Bernstein basis, to perform this conversion evaluate it using the de Casteljau algorithm, interpreting all barycentric combinations in this algorithm as multiplications by the functions:

$$u = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}, \quad (1 - u) = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$$

3.3 Composition

Suppose we are to compute the coefficients arising from composing two polynomials in the s-power basis. This is done efficiently taking into account that expression (3) admits a Horner's evaluation scheme:

$$a(u) = a_0 + s(a_1 + s(a_2 + s(a_3 + \cdots))), \qquad \begin{aligned} s &= (1 - u)u \\ a_k &= (1 - u)a_k^0 + ua_k^1 \end{aligned}$$

The composition $a(u(v))$, where $u(v)$, $v \in [0, 1]$ is another polynomial, reduces to replacement of the variable u with $u(v)$. Hence, we just need to carry out polynomial additions and multiplications to compute the coefficients of $a(u(v))$.

As discussed by DeRose [1988] and DeRose et al. [1993], composition can be employed to implement many CAGD operations, in particular to solve the subdivision problem. Given a polynomial curve in the s-power basis and a parameter value $\alpha \in (0, 1)$ splitting the domain into two intervals, we find the s-power form of the resulting curve segments I, II by simply introducing the following linear changes of variables $u(v)$:

$$\mathrm{I} : u \in [0, \alpha] \qquad u(v) = v\alpha = \begin{Bmatrix} 0 \\ \alpha \end{Bmatrix}$$

$$\mathrm{II} : u \in [\alpha, 1] \qquad u(v) = \alpha(1 - v) + v = \begin{Bmatrix} \alpha \\ 1 \end{Bmatrix}$$

3. 4 Differentiation and Integration

Assume we have a polynomial $a(u)$ of order p (degree $n = 2p + 1$) and we want to get the coefficients c_k corresponding to its derivative $c(u)$:

$$a(u) = \sum_{k=0}^{p} a_k s^k \xrightarrow{d/du} c(u) = \sum_{k=0}^{p} c_k s^k, \qquad s = (1 - u)u$$

To differentiate the kth term $a_k s^k$, we must take into account that both a_k and s are actually functions of u. Applying the product rule and adopting notation (4):

$$\frac{d}{du}(a_k s^k) = s^k \frac{da_k}{du} + a_k k s^{k-1} \frac{ds}{du} \qquad \begin{array}{l} \dfrac{da_k}{du} = \Delta a_k \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \\ \dfrac{ds}{du} = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \end{array}$$

Simplifying:

$$\frac{d}{du} a_k s^k = k \begin{Bmatrix} a^0 \\ -a^1 \end{Bmatrix}_k s^{k-1} + (2k+1)\Delta a_k s^k \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}. \tag{10}$$

Therefore, the derivative $c(u)$ is a polynomial of order p and coefficients

$$c_k = (2k+1)\Delta a_k \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} + (k+1) \begin{Bmatrix} a^0 \\ -a^1 \end{Bmatrix}_{k+1}, \quad k = 0, \ldots, p \tag{11}$$

By introducing decomposition (5) of the linear function $a_k(u)$ (4) into a constant and a skew-symmetric linear component, we can rewrite the differentiation rule (11) as

$$\begin{aligned} \Delta c_k &= -2(k+1)\bar{a}_{k+1} \\ \bar{c}_k &= (2k+1)\Delta a_k - \frac{k+1}{2}\Delta a_{k+1} \end{aligned} \tag{12}$$

This relation shows, not surprisingly, that the skew-symmetric term Δc_k comes from the symmetric component of $a(u)$, and vice versa. Note that in the case $k = p$ we have that $a_{p+1} = 0$, and therefore $\Delta c_p = 0$. Thus, as expected, the real degree of the derivative $c(u)$ is $(n - 1) = 2p$.

We now address the reverse problem: Given a polynomial $c(u)$ of order p, find its integral $a(u)$. Just solve the relationships (12) for a_k, $\bar{a}_k$ and compute them as follows:

$$\begin{aligned} \bar{a}_k &= -\frac{\Delta c_{k-1}}{2k} & k &= 1, \ldots, p+1 \\ \Delta a_k &= \frac{1}{2k+1}\left[\bar{c}_k + \frac{k+1}{2}\Delta a_{k+1}\right] & k &= p, \ldots, 0 \end{aligned}$$

where $\bar{a}_0$ is undetermined, because it corresponds to an arbitrary integration constant. The recursion formula for the values Δa_k starts with $\Delta a_{p+1} = 0$ and proceeds backwards. The requested coefficients a_k of the integral polynomial $a(u)$ are calculated using formula (5).

3.5 Division

The algorithm for dividing two polynomials in the s-power form is akin to "long division" for polynomials in power basis. First, write both polynomials with their terms in increasing order, then multiply the divisor by the appropriate monomial to eliminate the leading term of the dividend, subtract to obtain a remainder, and repeat the process. This is an example of an "online" algorithm [Knuth 1997]. It can be used to determine $k + 1$ coefficients $c_0, \ldots, c_k$ of the result without knowing k in advance, so it is possible in theory to run the algorithm indefinitely.

Assume we have two polynomials $a(u)$, $b(u)$, in s-power form, of coefficient lists **a**, **b**, respectively, such that $b_0^0 \neq 0$, $b_0^1 \neq 0$, and we want to compute the quotient $c(u) = a(u)/b(u)$. In general $c(u)$ cannot be expressed as a polynomial, yet we may calculate an approximation, the order-k Hermite interpolant $H_k(c;u)$. This is tantamount to decomposing the quotient $c(u)$ as

$$\frac{a(u)}{b(u)} = H_k(c;u) + \frac{r^{(k)}(u)}{b(u)}, \qquad H_k(c;u) = \sum_{i=0}^{k} c_i s^i, \tag{13}$$

where $r^{(k)}(u)$ denotes the kth remainder, whose terms are of order $> k$. Hence, $r^{(k)}(u)/b(u)$ does not contribute to the derivatives of order $\leq k$ at $u = \{0, 1\}$.

To begin with, the order-0 interpolant (7) is simply the linear function interpolating the values of $c(u)$ at the endpoints $u = \{0, 1\}$:

$$H_0(c;u) = c_0 = \begin{Bmatrix} a_0^0/b_0^0 \\ a_0^1/b_0^1 \end{Bmatrix}.$$

To obtain the coefficient c_k we first isolate the k-remainder from (13):

$$r^{(k)}(u) = a(u) - H_k(c;u)b(u) = a(u) - [H_{k-1}(c;u) + c_k s^k]b(u)$$

and express it as a function of the $(k - 1)$ remainder:

$$r^{(k)}(u) = r^{(k-1)}(u) - (c_k b(u))s^k. \tag{14}$$

All terms in the k-remainder (14) are of order $> k$, and hence its kth coefficient vanishes. Since the initial coefficient of the product $c_k b(u)$ coincides with the Hermite interpolant $H_0(c_k b_0;u)$, such a null kth coefficient is given by

$$0 = r_k^{(k-1)} - H_0(c_k b_0;u),$$

where $r_k^{(k-1)}$ denotes the kth term of the $(k - 1)$ remainder. Isolating c_k:

$$c_k = H_0\left(\frac{r_k^{(k-1)}}{b_0};u\right). \tag{15}$$

The resulting division algorithm is derived from expressions (14) and (15):

$c(u) := 0$
$r(u) := a(u)$
for $i := 0$ **to** k **do**
  $c_i := H_0(r_i / b_0;u)$
  $c(u) := c(u) + c_i s^i$
  $r(u); = r(u) - c_i s^i$
endfor

If we employ the data structure proposed in Section 3.2, rewriting the algorithm in terms of convolutions and vector shifts is a straightforward exercise.

3.6 Extracting Square Roots

Given a polynomial $a(u)$, the goal is to obtain the order-k Hermite approximation $H_k(c;u)$ of the function $c(u) = \sqrt{a(u)}$. For $k > 0$, we impose the additional constraint $a_0^0 > 0$, $a_0^1 > 0$ to ensure that $\sqrt{a(u)}$ is differentiable at $u = \{0, 1\}$.

Similarly to the division case, we split the result:

$$\sqrt{a(u)} = \sqrt{[H_k(c;u)]^2 + r^{(k)}(u)} \tag{16}$$

where the kth remainder $r^{(k)}(u)$ has terms of order $> k$ and, consequently, it does not contribute to the derivatives of order $\leq k$ at $u = \{0, 1\}$.

The initial coefficient c_0 represents the linear function interpolating the values of $\sqrt{a(u)}$ at $u = 0, 1$, that is, the order-0 Hermite interpolant:

$$c_0 = H_0(\sqrt{a(u)};u) = \begin{Bmatrix} \sqrt{a_0^0} \\ \sqrt{a_0^1} \end{Bmatrix}.$$

To compute the general coefficient c_k, we isolate the kth remainder from (16):

$$r^{(k)}(u) = a(u) - [H_k(c;u)]^2 = a(u) - [H_{k-1}(c;u) + c_k s^k]^2 \tag{17}$$

and expand the parenthesis:

$$r^{(k)}(u) = a(u) - [H_k(c;u)]^2 - [2H_{k-1}(c;u) + c_k s^k]c_k s^k.$$

Expressed as a function of the $(k-1)$th remainder:

$$r^{(k)}(u) = r^{(k-1)}(u) - [2H_{k-1}(c;u) + c_k s^k]c_k s^k. \tag{18}$$

All terms in the kth remainder (18) are of order $> k$. Hence, the kth term in expression (18) must be zero:

$$0 = r_k^{(k-1)} - 2H_0(c_0 c_k;u)$$

where $r_k^{(k-1)}$ denotes the kth term of the $(k-1)$ remainder. Isolating c_k:

$$c_k = \frac{1}{2}H_0\left(\frac{r_k^{(k-1)}}{c_0};u\right). \tag{19}$$

Equations (18) and (19) yield the following "online" algorithm:

$c(u) := 0$
$c_0 := H_0(\sqrt{a(u)};u)$
$r(u) := a(u) - c_0 c_0$
for $i := 1$ **to** k **do**
  $c_i := 1/2H_0(r_i/c_0;u)$
  $r(u) := r(u) - [2c(u) + c_i s^i]c_i s^i$
  $c(u) := c(u) + c_i s^i$
endfor

The algorithm to compute the Nth root is an immediate generalization, and only slightly more complex. We would expand a parenthesis raised to the Nth power in a formula analogous to (17) and follow a similar methodology.

4. S-POWER SERIES

4.1 Definition

For the two operations (division and square root) whose result $a(u)$ is not, in general, a polynomial with finite number of terms, we have already discussed how to compute as many terms as desired of a Hermite approximation. Formally speaking, we have defined an infinite series:

$$a(u) = \sum_{k=0}^{\infty} a_k s^k = (1-u)a^0(s) + ua^1(s), \qquad a^{0/1}(s) = \sum_{k=0}^{\infty} a_k^{0/1} s^k \tag{20}$$

whose truncation at the kth term (kth partial sum) yields the order-k Hermite interpolant $H_k(a;u)$ of $a(u)$. Such a series is referred to as a s-power series, and can be considered the two-point analogue of Taylor expansions. The s-power series is said to converge if as $k \to \infty$, the kth partial sum tends to the value of the function $a(u)$. In this section we explore the convergence of s-power series and how to employ them to approximate general functions $a(u)$.

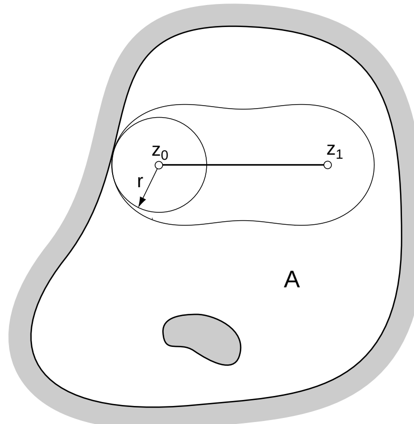

Fig. 3. Areas of convergence in the complex plane for a Taylor and a two-point Hermite series.

Equation (20) indicates that an s-power series is simply a linear combination of two standard power series $\alpha^{0/1}(s)$ in s. Hence it is convenient to summarize the fundamental results regarding the convergence of power series, which are easily formulated in the complex field [Silverman 1972]. Consider a power series of complex variable z:

$$a(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k \tag{21}$$

centered at a complex point z_0 and with complex coefficients a_k. Then there exists a nonnegative real number r, called the radius of convergence, such that:

- The series converges uniformly for those z verifying $|z - z_0| < r$, that is, interior to a disk of radius r and center z_0.
- The series diverges if $|z| > r$, that is, for points z exterior to the disk.

A function $a(z)$ differentiable in a disk centered at a point z_0 is called analytic (or regular) at z_0. If the function is not analytic at z_0, then we say that it presents a singularity at z_0. A function is analytic in a region A if it is analytic at each point of A. The interesting fact is that a function analytic at z_0 admits in a neighborhood of z_0 a representation as a series (21) with radius of convergence $r > 0$. Such a series is called a Taylor expansion, and its radius of convergence is precisely equal to that of the largest disk centred at z_0 we can draw in A, so that the disk does not contain singular points. This geometry is illustrated in Figure 3.

4.2 Lemniscates and the Convergence Theorem for Interpolatory Processes

Both the Taylor and two-point Hermite interpolations can be regarded as limiting cases of a more general formulation, that of finding the unique degree-n polynomial that interpolates to a given function $a(z)$ at $n + 1$ points z_i [Davis 1975]:

- If all points z_i tend to a common z_0, then we are prescribing derivatives at z_0 and we have a Taylor interpolation.
- If the points z_i tend to two distinct points z_0, z_1, we have our familiar two-point Hermite interpolation problem. In particular, we assumed up until now points $z_0 = 0$, $z_1 = 1$ on the real axis, but we could choose two arbitrary complex points.

Theorem 4.4.3 of Davis [1975] explains what occurs when $n \to \infty$ and the points to interpolate tend to K distinct points $z_0, z_1, \ldots, z_{K-1}$. In short, the convergence properties of the interpolating polynomial are those of Taylor expansions, replacing disks of radius r and center z_0 by lemniscates of radius r and K foci. For our case $K = 2$, given a function $a(z)$ analytical in a region A and two points $z_0, z_1 \in A$, as $n \to \infty$ the corresponding Hermite polynomial converges uniformly to $a(z)$ within the largest lemniscate of foci z_0, z_1 we can fit in A (Figure 3), and diverges outside. A lemniscate of foci z_0, z_1 is the locus of points z that move in such a way that the product of their distances to z_0 and z_1 is a constant r^2 [Davis 1975]:

$$|(z - z_0)(z - z_1)| = r^2, \quad r > 0 \tag{22}$$

Figure 4 shows that, as r varies, we have confocal lemniscates belonging to three different families:

- $0 < r < 1/2$: Two mutually exterior ovals, called Ovals of Cassini, one surrounding z_0 and the other z_1. For very small r, these ovals degenerate to circles.
- $r = 1/2$: A lemniscate of Bernoulli, an ∞-shaped figure with a double point at $(z_0 + z_1)/2$.
- $r > 1/2$: A closed contour containing both foci. For very large r, this figure degenerates to a circle.

4.3 Convergence of s-Power Series Over [0,1]

Now we turn back to our original problem, the convergence of s-power series for real functions $a(u)$ defined over the unit interval [0,1] of the real axis. In this particular case, $[z_0, z_1] = [0, 1]$, Eq. (22) of a lemniscate $\mathscr{L}_r$ of radius r becomes

$$|s(z)| = r^2, \qquad s(z) = (1 - z)z. \tag{23}$$

For a function defined over the unit interval [0,1], we just need convergence over [0,1], that is, the segment [0,1] must be contained in the largest

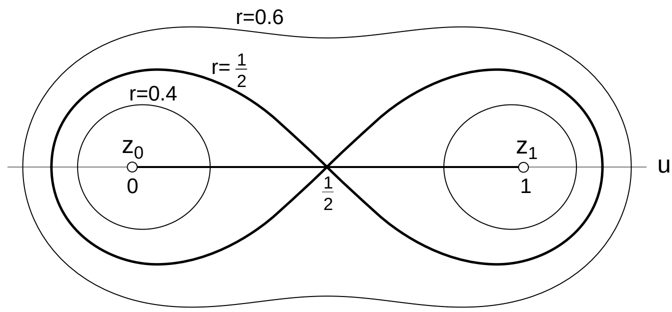

Fig. 4. Families of confocal lemniscates for increasing radii r.

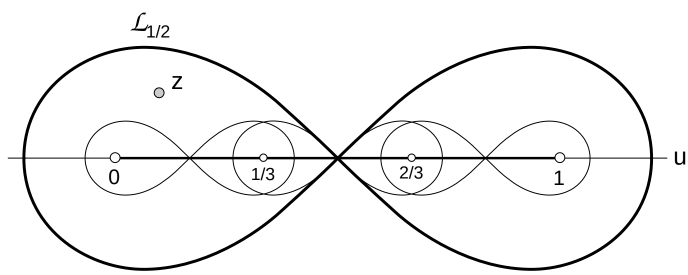

Fig. 5. Limiting lemniscates of Bernoulli $\mathscr{L}_{1/2}$ in the complex plane after subdivision of the unit parameter [0,1].

lemniscate $\mathscr{L}_r$ of foci $z = 0$, $z_1 = 1$ we can draw that does not contain singularities (Figure 4). So it must be $r > 1/2$ or, equivalently, a lemniscate $\mathscr{L}_{1/2}$ of Bernoulli with foci 0, 1 must not contain singularities. By (23) the s-power series enjoys convergence over [0,1] if all singular points z of $a(u)$ verify

$$|s(z)| > 1/4, \quad s(z) = (1 - z)z. \tag{24}$$

As the mapping $z \rightarrow s(z)$ transforms lemniscates $\mathscr{L}_r$ of foci 0,1 to circles of radius r centered at the origin $z_0 = 0$, condition (24) is tantamount to saying that the images $s(z)$ of all singularities fall outside the disk of radius $r = 1/4$ centered at $z_0 = 0$. Intuitively, such singularities must lie far away, so that they do not influence the behavior of $a(u)$ over [0,1].

If a singularity z lies on [0,1], there is no way to circumvent it. Nevertheless, in all other cases we can always get convergent s-power series through subdivision. This is shown in the example of Figure 5, where the unit interval has been subdivided into three equal segments. The corresponding limiting lemniscates of Bernoulli $\mathscr{L}_{1/2}$ for each of them are smaller and completely contained in the original one. Hence, it is easily understood that after successive subdivisions any singularity can be avoided.

The difference

$$R_k(a;u) = a(u) - H_k(a;u)$$

is called the remainder and measures the precision with which $a(u)$ is approximated by $H_k(a;u)$ at u. For example, in the case of s-power series arising from a polynomial division, this remainder coincides with the term $r^{(k)}(u)/b(u)$ of Eq. (13). This remainder is characterized as a particular case of Theorem 3.5.1 in Davis [1975]:

$$R_k(a;u) = \frac{a^{(n+1)}(\xi)}{(n+1)!}(-s)^{k+1}, \qquad n = 2k + 1$$

where ξ is a suitable, not specified, intermediate parameter value $\xi \in (0, 1)$. This formula, called Cauchy's remainder, applies to any function $a(u)$ enjoying continuity within [0,1] and such that the derivative $a^{(n+1)}$ exists in (0,1). Observe that a necessary and sufficient condition for the convergence of an s-power series at a given point u is that $R_k(a;u) \to 0$ as $k \to \infty$.

4.4 Division Revisited

With this theoretical background we can now turn back to explore the convergence of the s-power series corresponding to the quotient $c(u) = a(u)/b(u)$ of two polynomials. Since the only singularities of $c(u)$ are the roots of $b(u)$ not contained in $a(u)$, then if such roots verify condition (24), the s-power series converges over [0,1]. It is worth mentioning that this s-power series can be expressed in terms of a standard power series if we multiply and divide by $b(1-u)$:

$$c(u) = \frac{a(u)}{b(u)} = a(u)b(1-u)\frac{1}{\hat{b}(s)}, \qquad \hat{b}(s) = b(u)b(1-u). \tag{25}$$

According to (6), the denominator $\hat{b}(s)$ is a true polynomial in monomial form in s (whose coefficients are constants). Equation (25) indicates that the s-power series of $c(u)$, with $u \in [0, 1]$, inherits the convergence properties from the power expansion of $1/\hat{b}(s)$, $s \in [0, 1/4]$. As a root z of $b(u)$ transforms to a root $s(z)$ of $\hat{b}(s)$, we again reach the condition of convergence (24).

Now consider the following simple example:

$$c(u) = \frac{1}{z(u)}, \qquad z(u) = \begin{Bmatrix} z^0 \\ z^1 \end{Bmatrix} \tag{26}$$

where $z(u)$ denotes a linear function in Bernstein form (4). Such a function $z(u)$ has a single real root u_0. Hence, condition (24) of convergence, i.e., that u_0 lies outside $\mathscr{L}_{1/2}$, can be rewritten more explicitly as

$$u_0 \notin [\mu, 1-\mu], \qquad \begin{aligned} u_0 &= \frac{-z^0}{\Delta z} \\ \mu &= \frac{1}{2} - \frac{\sqrt{2}}{2}. \end{aligned} \tag{27}$$

After some algebraic manipulation, Eq. (25) becomes

$$c(u) = \begin{Bmatrix} 1/z^0 \\ 1/z^1 \end{Bmatrix} \frac{1}{1 - s/s_0}, \qquad s_0 = s(u_0) = -\frac{z^0 z^1}{(\Delta z)^2}$$

The power series of $1 - s/s_0$ derives from the geometrical series:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots \qquad |x| < 1 \tag{28}$$

by substituting $x = s/s_0$; note that the condition $|x| < 1$ for the convergence of the geometrical series (28) is equivalent to (24). Thus, the general kth coefficient c_k of $c(u)$ (26) admits the following closed-form expression:

$$c_k = \begin{Bmatrix} 1/z^0 \\ 1/z^1 \end{Bmatrix} (s_0)^{-k}. \tag{29}$$

Figure 6(a) shows in the case $z(u) = \begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$ how the successive order-k partial sums rapidly converge within $u \in [0, 1]$, because $u_0 = -1$, and so condition (27) is fulfilled. In the case $z(u) = \begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$ plotted in Figure 6(b), we have a central root $u_0 = 1/2$ not verifying condition (27), where $c(u)$ has a singularity. However, the s-power series converges to $c(u)$ for all other points over [0,1].

4.5 Some Remarkable s-Power Series

All well-known techniques to obtain the standard power series of a given function $a(u)$ can also be applied to compute the corresponding s-power series, in particular, the method of undetermined coefficients [Courant and John 1989]. To begin with, we write down tentatively $a(u) = \sum_{k=0}^{\infty} a_k s^k$, where the coefficients a_k are in principle unknown. Then we determine the coefficients by some known property of $a(u)$.

For example, assume we want to get the s-power series of the exponential function $a(u) = e^{z(u)}$, where $z(u)$ is a linear function of Bézier coefficients $z^{0/1}$. Since this function verifies the differential equation:

$$a(u)\Delta = \frac{d}{du} a(u), \quad \Delta = \frac{d}{du} z(u) = z^1 - z^0$$

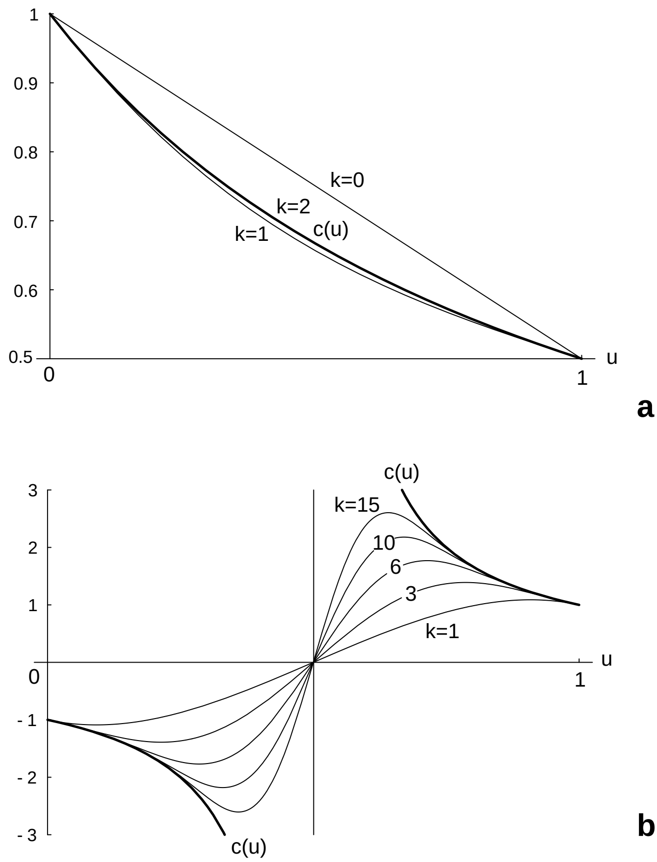

Fig. 6. Successive order-k two-point Hermite expansions for the inverse $c(u)$ of a linear function $z(u)$ with Bézier coefficients: (a) $\begin{Bmatrix} 1 \\ 2 \end{Bmatrix}$; (b) $\begin{Bmatrix} -1 \\ 1 \end{Bmatrix}$.

we just replace c_k with $a_k\Delta$ in Eq. (11) and solve for a_{k+1}. In matrix form:

$$a_{k+1} = \begin{Bmatrix} a^0 \\ a^1 \end{Bmatrix}_{k+1} = \frac{1}{k+1}\begin{bmatrix} n+\Delta & -n \\ -n & n-\Delta \end{bmatrix}\begin{Bmatrix} a^0 \\ a^1 \end{Bmatrix}_k, \quad n = 2k+1. \qquad (30)$$

Starting with $a_0 = H_0(a;u)$, the recursive formula (30) for $k = 0, 1, 2, \ldots$ generates the successive coefficients of the requested s-power series. Such a series enjoys convergence over [0,1] because $a(u)$ has no singularities.

The same methodology holds for generating the s-power series of $a(u) = [z(u)]^\alpha$, where $z(u)$ is a linear function of coefficients $z^{0/1}$ and α is any real number. The function $a(u)$ satisfies the relation

$$a(u)\alpha\Delta = z(u)\frac{d}{du}a(u), \quad \Delta = z^1 - z^0 \qquad (31)$$

The right side is obtained by multiplying term (10) by the linear function $z(u)$. After simplification:

$$\begin{Bmatrix} z^0 \\ z^1 \end{Bmatrix} \frac{d}{du} a_k s^k = k \begin{Bmatrix} a_k^0 z^0 \\ -a_k^1 z^1 \end{Bmatrix} s^{k-1} + \begin{Bmatrix} (k\Delta - nz^0)a_k^0 + (k\Delta + nz^0)a_k^1 \\ (k\Delta - nz^1)a_k^1 + (k\Delta + nz^1)a_k^1 \end{Bmatrix} s^k, \quad n = 2k+1$$

Equating terms in relation (31):

$$a_k \alpha \Delta = (k+1) \begin{Bmatrix} a_{k+1}^0 z^0 \\ -a_{k+1}^1 z^1 \end{Bmatrix} + \begin{Bmatrix} (k\Delta - nz^0)a_k^0 + (k\Delta + nz^0)a_k^1 \\ (k\Delta - nz^1)a_k^0 + (k\Delta + nz^1)a_k^1 \end{Bmatrix}, \quad n = 2k+1$$

Now we solve for a_{k+1} to get the recursive formula:

$$\begin{Bmatrix} a^0 \\ a^1 \end{Bmatrix}_{k+1} = \frac{1}{k+1} \begin{bmatrix} n + (\alpha - k)\dfrac{\Delta}{z^0} & -n - k\dfrac{\Delta}{z^0} \\ -n + k\dfrac{\Delta}{z^1} & n + (k - \alpha)\dfrac{\Delta}{z^1} \end{bmatrix} \begin{Bmatrix} a^0 \\ a^1 \end{Bmatrix}_k \quad k = 0, 1, 2 \ldots$$

and start with $a_0 = H_0(a;u)$. This s-power series exhibits convergence over [0,1] for any real value α if the root u_0 of $z(u)$ satisfies condition (27).

We can also generate, through integration, the s-power series for any function $a(u)$ whose derivative $c(u)$ involves only quotients or square roots of polynomials. As the s-power series for $c(u)$ is computable, it suffices to isolate a_{k+1} in Eq. (11):

$$a_{k+1} = \begin{Bmatrix} a^0 \\ a^1 \end{Bmatrix}_{k+1} = \frac{1}{k+1} \begin{Bmatrix} c_k^0 - (2k+1)\Delta a_k \\ -c_k^1 + (2k+1)\Delta a_k \end{Bmatrix}, \quad k = 0, 1, \ldots$$

This recursive formula starts with $a_0 = H_0(a;u)$ and proceeds forwards. For example, we generate the s-power series $a(u)$ for the natural logarithm $\ln(z(u))$ of a linear function $z(u)$ by integrating $c(u) = \Delta / z(u)$, where $\Delta = z^1 - z^0$. The coefficients c_k are simply those of expression (29) multiplied by Δ. This method also applies to the inverses of trigonometric and hyperbolic functions.

5. EXAMPLES OF THE APPLICATION

5.1 Integral Approximation of Rational Curves

A common problem that arises when trying to exchange data between dissimilar CAD-CAM systems is finding an integral, i.e., polynomial approximation of a rational curve. If a system that can only handle integral curves has to import a rational curve, then the latter must be approximated. If we transform a given rational curve to the s-power basis, we can compute its polynomial Hermite approximation in a trivial way: we just apply the 7-line division algorithm of Section 3.5. Figure 7(a) shows an

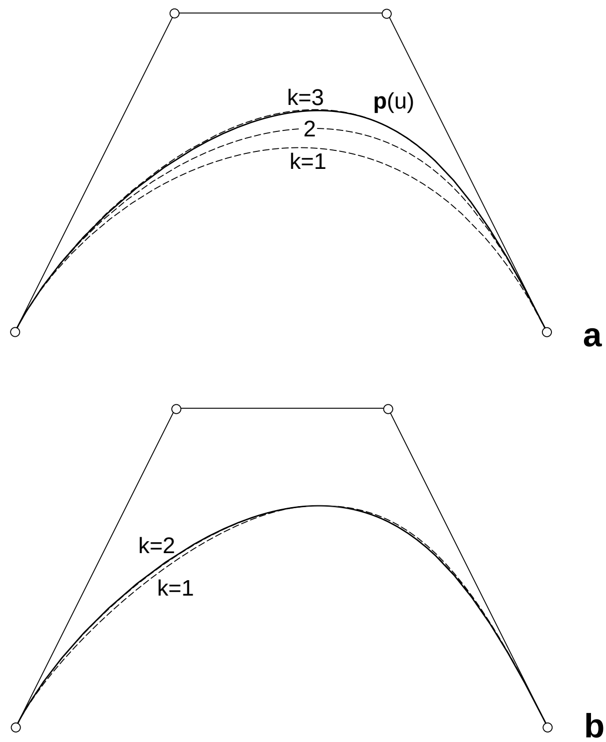

Fig. 7. Successive order-k Hermite approximations of a cubic rational curve $\mathbf{p}(u)$.

example where we have plotted successive order-k integral approximations of a cubic rational curve $\mathbf{p}(u)$ defined in Bézier form, with weights {1, 1/2, 1, 1}. Such approximations converge to the rational curve because the roots of the denominator fulfil condition (24). In Figure 7(b) we first subdivided $\mathbf{p}(u)$ at the midpoint $u = 1/2$ and then computed the order-k approximations for each of the two segments. Such approximations form a C^k spline that clearly improves the result. Observe that the order-2 approximations are visually indistinguishable from $\mathbf{p}(u)$. In case the denominator is contained as a multiplying factor in the numerator, there exists an exact solution, which the Hermite interpolant recaptures. This is the case of a rational curve obtained from a polynomial one by inserting base points [Farin 1995].

The proposed Hermite interpolant coincides with a particular instance of a "hybrid Bézier curve," as pointed out by Sederberg and Kakimoto [1991] and hinted by Eq. (13). If the numerator and denominator are of degree at most n, the remainder $r^{(k)}(u)$ (14) has degree at most $n + 2k + 1$, and hence can be expressed as

$$r^{(k)}(u) = s^m d(u) = B_m^{2m}(u)\binom{2m}{m}^{-1} d(u), \qquad m = k + 1$$

where $d(u)$ denotes a certain degree-n polynomial and $B_m^{2m}(u)$ is the middle Bernstein polynomial of degree $2m$. This leads to rewriting decomposition (13) as a "hybrid polynomial" of degree $2m$ whose central Bézier ordinate (index m) is not a constant but a rational polynomial of degree n. It is worth stressing that, if the Bernstein basis is utilized, the computation of the Hermite interpolant involves heavy algebra, and the same holds true for the convergence condition [Wang and Zheng 1997].

5.2 Hermite Approximation of Offsets to Parametric Curves

Offset curves are of interest because they arise in the mathematical description of many operations, including the description of milling processes [Hoschek and Lasser 1993], the representation of thick planes or path planning for rapid prototyping. The (untrimmed) offset $\mathbf{p}_r(u)$ to a given plane curve $\mathbf{p}(u) = (x(u), y(u))$ at distance r is the curve

$$\mathbf{p}_r(u) = \mathbf{p}(u) + r\mathbf{n}(u), \quad \mathbf{n}(u) = \frac{(-y'(u), x'(u))}{\sqrt{x'^2(u) + y'^2(u)}} \tag{32}$$

where $\mathbf{n}(u)$ denotes the unit normal at the point $\mathbf{p}(u)$ for a given orientation of the curve.

Despite its apparent simplicity, offsetting is one of the most difficult geometric operations to implement. As widely known, the set of rational curves $\mathbf{p}(u)$

$$\mathbf{p}(u) = (x(u), y(u)) = \left(\frac{a_x(u)}{w(u)}, \frac{a_y(u)}{w(u)}\right) \tag{33}$$

is not closed under the offsetting operation. In other words, the offset to a rational curve $\mathbf{p}(u)$ (33) cannot in general be expressed exactly as a rational curve, due to a square root function in the expression of the normal $\mathbf{n}$ (32) to a curve (33):

$$\mathbf{n}(u) = \frac{(d_x(u), d_y(u))}{\sqrt{d_x^2(u) + d_y^2(u)}} \qquad \begin{aligned} d_x(u) &= -a'_y(u)w(u) + w'(u)a_y(u) \\ d_y(u) &= a'_x(u)w(u) - w'(u)a_x(u) \end{aligned} \tag{34}$$

Therefore, exact offsets are confined to special subsets of curves [Lü 1995; Pottmann 1995]. In particular, a sufficient condition for the offset of an integral curve to be rational is that its hodograph (derivative) $(x'(u), y'(u))$ have polynomial norm. Those planar curves exhibiting this property are called Pythagorean-hodograph (PH) curves [Farouki 1992 ; 1994].

Approximation techniques [Pham 1992 ; Elber et al. 1997] seem to be the only feasible practical solution to the general offsetting problem. Again, we advocate computing the Hermite approximation of the offset, as proposed by Sederberg and Buehler [1992]. Whereas this task turns out to be very involved using the customary Bernstein basis, due to the difficulty of computing convenient approximations for the square root and division in (34), it is trivial in the s-power basis. To obtain the order-k Hermite approximation of the offset, simply compute order-k Hermite approxima-

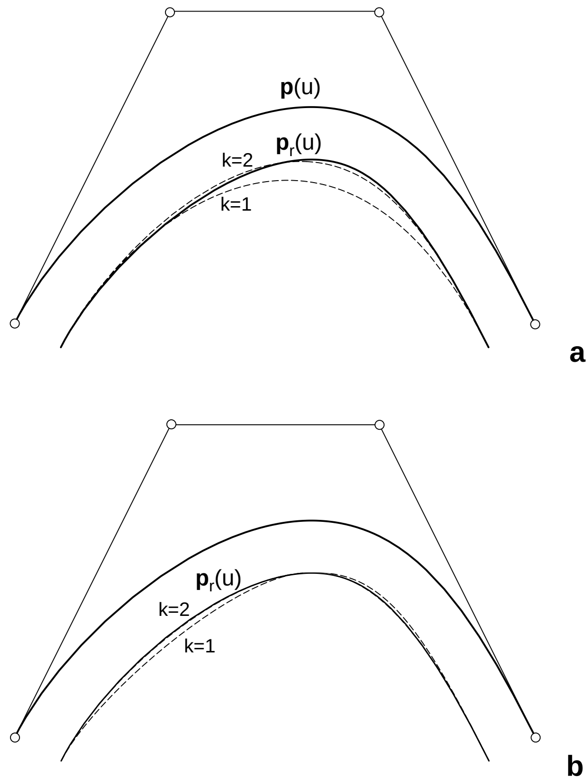

Fig. 8. Successive order-k Hermite approximations to the offset $\mathbf{p}_r(u)$ of a cubic curve $\mathbf{p}(u)$.

tions for $\mathbf{p}(u)$ and $\mathbf{n}(u)$ (34). This has been done in Figure 8(a) to compute the offset of the same rational curve $\mathbf{p}(u)$ previously used in Figure 7. The condition of convergence to the offset as $k \to \infty$ is that the singularities of both $\mathbf{p}(u)$ and $\mathbf{n}(u)$, the zeros z of $w(u)$ (33), and the denominator in (34) verify condition (24). In Figure 8(b) we first subdivided $\mathbf{p}(u)$ at the midpoint $u = 1/2$ and then plotted the approximations for the two resulting segments.

We could also generate a rational approximation, rather than a polynomial one, if we compute an order-k Hermite approximation only for the square root in (34). Thus when we take a Pythagorean-hodograph curve of degree n, whose offset is a rational curve of degree at most $2n - 1$, this method recaptures the exact offset if we choose k large enough, so that $2k + 1 \geq 2n - 1$.

6. EXTENSION TO BIVARIATE POLYNOMIALS

6.1 Bivariate Polynomials in the Symmetric Power Form

Most concepts and definitions of the univariate case carry over naturally to the bivariate case. A bivariate polynomial $a(u, v)$ of degree $(m, n) = (2p$

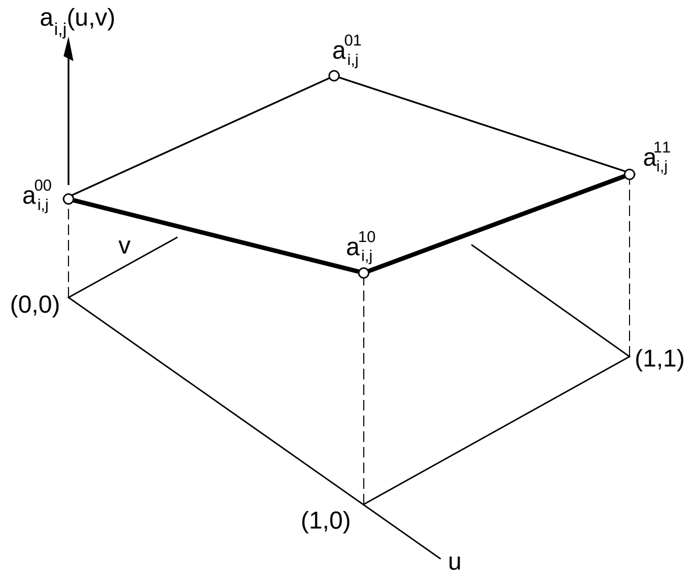

Fig. 9. Interpreting $a_{i,j}(u, v)$ as a bilinear function in Bézier form.

$+ 1, 2q + 1)$ over the domain $(u, v) \in [0, 1] \times [0, 1]$ is written in the s-power basis as

$$a(u, v) = \sum_{i=0}^{p} \sum_{j=0}^{q} a_{i,j}(u, v) s^i t^j, \qquad \begin{array}{l} s = (1 - u)u \\ t = (1 - v)v \end{array} \tag{35}$$

where the coefficients

$$a_{i,j}(u, v) = \{(1 - u), u\} \begin{bmatrix} a^{00} & a^{01} \\ a^{10} & a^{11} \end{bmatrix}_{i,j} \begin{Bmatrix} 1 - v \\ v \end{Bmatrix}$$

are not constants, but bilinear functions. For algebraic manipulation purposes, such a bilinear function is defined by the four values corresponding to its Bézier representation (Figure 9), and is represented by a 2×2 matrix:

$$a_{i,j}(u, v) \rightarrow a_{i,j} = \begin{bmatrix} a^{00} & a^{01} \\ a^{10} & a^{11} \end{bmatrix}_{i,j}. \tag{36}$$

Degree-related operations in this bivariate basis become trivial. A truncation of the sum (35) at the (k, l)th term yields the order-(k, l) Hermite interpolant that matches all derivatives

$$\frac{\partial a(u, v)}{\partial^i u \partial^j v} \qquad \begin{array}{l} i \leq k \\ j \leq l \end{array}$$

at the four corners of the domain. In other words, this operation preserves as many derivatives as possible of the original polynomial $a(u, v)$ at the four corners. Note that a boundary is a polynomial curve whose coefficients in the s-power form are given by rows or columns of matrices (36) with a null subscript. For instance, the boundary $u = 0$ is determined by rows $\{a^{00}, a^{01}\}_{0,j}$, $j = 0, \ldots, q$. Hence, at each border we are clearly reproducing the degree-reduction scheme for curves (Section 2.4).

The order-0 Hermite interpolant $H_{0,0}(a;u, v)$ of a given function $a(u, v)$ reduces to the bilinear function interpolating $a(u, v)$ at the four corner points. Such a Hermite interpolant is represented as

$$H_{0,0}(a;u, v) \rightarrow \begin{bmatrix} a(0,0) & a(0,1) \\ a(1,0) & a(1,1) \end{bmatrix}.$$

6.2 Addition and Multiplication

Similarly to the univariate case, adding two bivariate polynomials in the s-power basis is simply a component-wise addition of their respective matrices (36).

As for multiplication, first consider that multiplying two polynomials with coefficient matrices $\mathbf{a}$, $\mathbf{b}$ in the standard power form reduces to computing the 2D analogue of the convolution (8):

$$\mathbf{c} = \sum_{i,j} \mathrm{shift}_{i,j}(b_{i,j}\,\mathbf{a})$$

We now initialize to zero a matrix $\mathbf{c}$ and add to it, starting at the (i, j)th position, submatrices $b_{i,j}\mathbf{a}$.

To multiply in the s-power form (35), we must know how to compute the product between two bilinear functions, represented by their respective matrices a, b (36). After some algebra:

$$ab = \begin{bmatrix} a^{00}b^{00} & a^{01}b^{01} \\ a^{10}b^{10} & a^{11}b^{11} \end{bmatrix} - s\begin{bmatrix} a_u^0 b_u^0 & a_u^1 b_u^1 \\ a_u^0 b_u^0 & a_u^1 b_u^1 \end{bmatrix} - t\begin{bmatrix} a_v^0 b_v^0 & a_v^0 b_v^0 \\ a_v^1 b_v^1 & a_v^1 b_v^1 \end{bmatrix}$$
$$+ st(a_u^1 - a_u^0)(b_u^1 - b_u^0)\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \tag{37}$$

where

$$\begin{aligned} a_u^0 &= a^{10} - a^{00} & \quad a_v^0 &= a^{01} - a^{00} \\ a_u^1 &= a^{11} - a^{01} & \quad a_v^1 &= a^{11} - a^{10} \end{aligned}$$

denote the partial derivatives of $a(u, v)$ at the corners, and analogously for $b(u, v)$. Notice that, in addition to the bilinear term whose representation is obtained by multiplying component by component, we get higher order terms.

To represent a polynomial $a(u, v)$, we use a 2×2 matrix (36) in which each coefficient is a $(p + 1) \times (q + 1)$ matrix:

$$a(u, v) \rightarrow \mathbf{a} = \begin{bmatrix} \mathbf{a}^{00} & \mathbf{a}^{01} \\ \mathbf{a}^{10} & \mathbf{a}^{11} \end{bmatrix}, \quad \mathbf{a}^{00} = \begin{bmatrix} a^{00}_{0,0} & \cdots & a^{00}_{0,q} \\ \vdots & a^{00}_{i,j} & \vdots \\ a^{00}_{p,0} & \cdots & a^{00}_{p,q} \end{bmatrix}.$$

Hence, the matrix $\mathbf{c}$ corresponding to the product $c(u, v) = a(u, v)\, b(u, v)$ is obtained from the product rule (37), replacing

coefficient	$\rightarrow$	coefficient matrix
product of coefficients	$\rightarrow$	2D-convolution* of matrices
multiplication by s	$\rightarrow$	S : shift a matrix 1 row position
multiplication by t	$\rightarrow$	T : shift a matrix 1 column position

Therefore, the component $\mathbf{c}^{00}$ is given by

$$\mathbf{c}^{00} = \mathbf{a}^{00} * \mathbf{b}^{00} - S(\mathbf{a}^0_u * \mathbf{b}^0_u) - T(\mathbf{a}^0_v * \mathbf{b}^0_v) + ST((\mathbf{a}^1_u - \mathbf{a}^0_u) * (\mathbf{b}^1_u - \mathbf{b}^0_u))$$

and analogously for $\mathbf{c}^{01}$, $\mathbf{c}^{10}$, $\mathbf{c}^{11}$.

6.3 Other Arithmetic Operations

Once we know how to multiply two bivariate polynomials in the symmetric power form, the algorithms for all other operations are straightforward generalizations of those for the univariate case. Thus, for the sake of conciseness, we develop division as an example only.

Given two bivariate polynomials $a(u, v)$, $b(u, v)$ in s-power form such that all terms in the 2 x 2 matrix b_{00} are nonzero, the goal is obtaining the order-(k, l) Hermite approximation $H_{k,l}(a; u, v)$ of the quotient $c(u, v)$:

$$\frac{a(u, v)}{b(u, v)} = H_{k,l}(c; u, v) + \frac{r^{(k,l)}(u, v)}{b(u, v)}, \qquad H_{k,l}(c; u, v) = \sum_{i=0}^{k} \sum_{j=0}^{l} c_{i,j} s^i t^j$$

In complete analogy to the univariate case, in the (k, l)th remainder $r^{(k,l)}(u, v)$ all terms have orders higher than (k, l), and the order-0 interpolant $c_{0,0}$ is the bilinear function interpolating the values of $c(u, v)$ at the corner points. We write both polynomials $a(u, v)$, $b(u, v)$ with their terms in increasing order, then multiply the divisor by the appropriate value $c_{i,j} s^i s^j$ to eliminate the leading term of the dividend, subtract to obtain a remainder, and repeat the process. The corresponding algorithm, now with two nested loops, is

$c(u, v) := 0$
$r(u, v) := a(u, v)$
for $i := 0$ **to** k **do**

for $j := 0$ **to** l **do**
  $c_{i,j} := H_{0,0}(r_{i,j} / b_{0,0}; u, v)$
  $c(u, v) := c(u, v) + c_{i,j} s^i s^j$
  $r(u, v) := r(u, v) - c_{i,j} s^i s^j b(u, v)$
endfor
endfor.

7. CONCLUSIONS

We presented a unified framework for processing polynomials, based on representing them in the symmetric analogue of the power basis (s-power basis) and algebraically computing all operations needed in this representation. The s-power form of a polynomial can be regarded as a Hermite two-point expansion over the interval $u \in [0, 1]$ into increasing degree terms. Therefore, a truncation at the kth term is tantamount to degree-reduction, yielding the order-k Hermite interpolant that reproduces as many derivatives as possible at the endpoints. Granted, the Hermite approximation is not the best possible in the sense of minimizing the error. However, it yields reasonable approximations well suited for the piecewise representation of curves and surfaces.

A polynomial in s-power form can be viewed as a power series of the parameter $s = (1 - u)u$ whose coefficients are linear functions, rather than constant values. As a consequence, all algorithms for algebraic manipulation, needed in geometry processing, are easily derived from those corresponding to the standard power series. Multiplication is easily expressed in terms of convolutions, and there exists a very simple online algorithm to compute as many terms as desired of the Hermite approximation of a quotient or square root. Thus, geometry processing operations such as computing an integral approximation of a rational curve or an approximation of an offset reduces to trivial tasks. The convergence properties of the resulting s-power series are akin to those of Taylor series, replacing disks with lemniscates in the complex plane. Bivariate polynomials over the interval $(u, v) \in [0, 1] \times [0, 1]$ in s-power form can be interpreted as power polynomials of parameter $(s, t) = ((1 - u)u, (1 - v)v)$, whose coefficients are bilinear functions.

In summary, we advocate using the s-power basis for geometry processing operations, as it combines the most positive properties of the Bernstein, Taylor, and Hermite polynomial bases, without their disadvantages:

- Like the power basis, it enjoys permanence, allowing the expansion of nonpolynomial functions in an infinite series.
- Similarly to the Bernstein basis, it enjoys good numerical properties, and the coefficients in the basis admit a geometric interpretation [Sánchez-Reyes 1997].
- It is adequate for a piecewise representation of curves and surfaces.

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